

# Rabbits, Traits and Natural Selection

PhET Natural Selection · PhET

Year 10

~30 minutes

Science · Biology

## NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/natural-selection>

Curriculum links: AC9S10U02 · evolution · natural selection · genetics

## Learning intentions

- I can explain how an environmental pressure changes which traits survive in a population.
- I can use generation-by-generation data to describe a shift in allele frequency.
- I can evaluate this simulation as evidence for the theory of evolution by natural selection.

## Getting started

Open PhET Natural Selection. Start on the Intro screen with the default rabbit population, then add a trait (e.g. fur colour) before switching on an environmental factor such as wolves or food scarcity.

## Tasks

1. Run the simulation for 10 generations with no environmental pressures switched on. What happens to the population size and mix of traits?

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2. Add wolves as a predator to a mixed-colour rabbit population (e.g. white and brown fur) on a plain background. Predict which fur colour will survive better, and why.

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3. Run the simulation for 10 generations with wolves active. Record the approximate percentage of each fur colour every 2 generations using the population graph.

| Generation | % brown fur | % white fur |
|------------|-------------|-------------|
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**4. Explain the pattern you recorded in your table. Why did one fur colour become more common than the other?**

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**5. Change the environment (e.g. switch the background to snow, or add food scarcity instead of wolves). Predict how this change might affect which trait is favoured, then test it.**

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**6. Using your results, explain what is meant by 'survival of the fittest' — is fitness about being the strongest rabbit, or something else? Use evidence from the simulation.**

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**7. Evaluate: does this simulation on its own prove that natural selection can produce a brand-new species? Explain what it does show, and what you would still need to check.**

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***Extension (optional)***

Antibiotic resistance in bacteria works on the same underlying mechanism as the rabbits in this simulation. Explain the similarity, including what plays the role of the 'environmental pressure' in each case.

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